2) Floods

Flood is an event in which a part of the earth's surface gets inundated. Heavy rainfall and large waves in seas are the common causes of flood.



Flood

The major causes of floods are:

A. Meteorological factors

i) Heavy rainfall

ii) Tropical cyclones

iii) Cloud burst

B. Physical factors

i) Large catchment area

ii) Inadequate drainage arrangement

C. Human factors

i) Deforestation

ii) Siltation

iii) Faulty agricultural practices

iv) Faulty irrigation practices

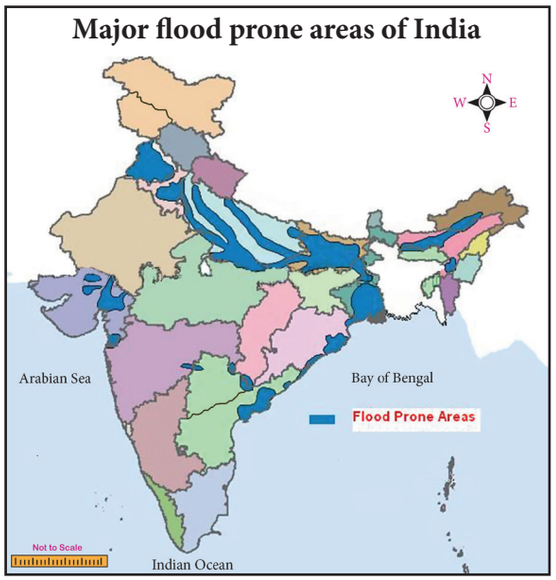
v) Collapse of dams

vi) Accelerated urbanisation

ACTIVITY

Discuss in the classroom about the actions to be taken before, during and after flood.

The following map shows the major flood prone areas in India. Gangetic plains covering the states of Punjab, Haryana, Uttar Pradesh, North Bihar, West Bengal and Brahmaputra valley are the major flood prone areas in north and northeast India. Coastal Andhra Pradesh, Odisha and southern Gujarat are the other regions which are also prone to flood often.



(Source: National Institute of Hydrology, New Delhi)

3) Cyclonic Storms

A cyclonic storm is a strong wind circulating around a low pressure area in the atmosphere. It rotates in anti-clockwise direction in Northern Hemisphere and clockwise in the Southern Hemisphere.

Tropical cyclones are characterised by destructive winds, storm surges and exceptional levels of rainfall, which may cause flooding. Wind speed may reach upto 200 km/h and rainfall may record upto 50 cm/day for several consecutive days.

A sudden rise of seawater due to tropical cyclone is called storm surge. It is more common in the regions of shallow coastal water.

East coastal areas vulnerable to storm surges

i) North Odisha and West Bengal coasts.

ii) Andhra Pradesh coast between Ongole and Machilipatnam.

iii) Tamil Nadu coast (among 14 coastal districts, Nagapattinam and Cuddalore districts are frequently affected).

West coastal areas vulnerable to storm surges

The west coast of India is less vulnerable to storm surges than the east coast.

i) Maharashtra coast, north of Harnai and adjoining south Gujarat coast and the coastal belt around the Gulf of Cambay.

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